Logistic Regression

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# Logistic Regression

from sklearn.linear_model import LogisticRegression

# Training data
# X = Hours studied
X = [[1], [2], [3], [4], [5], [6], [7], [8]]

# Y = Result
# 0 = Fail, 1 = Pass
y = [0, 0, 0, 0, 1, 1, 1, 1]

# Create the model
model = LogisticRegression()

# Train the model
model.fit(X, y)

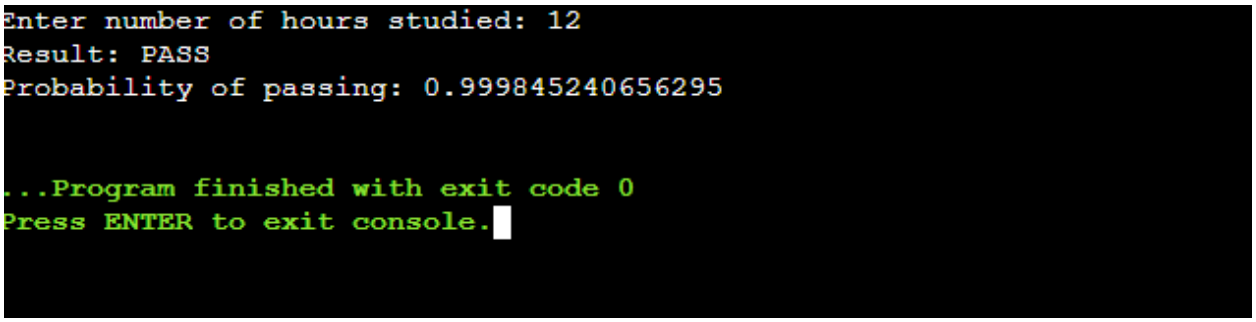
# Take input from user
hours = float(input("Enter number of hours studied: "))

# Predict result
prediction = model.predict([[hours]])
probability = model.predict_proba([[hours]])

# Display result
if prediction[0] == 1:
   print("Result: PASS")
else:
   print("Result: FAIL")

print("Probability of passing:", probability[0][1])
```

Output Of the Code:-


Enter number of hours studied: 12
Result: PASS
Probability of passing: 0.999845240656295
...Program finished with exit code 0
Press ENTER to exit console.